

Pragmatic inference of scalar implicature by LLMs

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Abstract

This study investigates how Large Language Models (LLMs), particularly BERT (Devlin et al., 2019) and GPT-2 (Radford et al., 2019), engage in pragmatic inference of scalar implicature, such as *some*. Two sets of experiments were conducted using cosine similarity and next sentence/token prediction as experimental methods. The results in experiment 1 showed that, both models interpret *some* as pragmatic implicature *not all* in the absence of context, aligning with human language processing. In experiment 2, in which Question Under Discussion (QUD) was presented as a contextual cue, BERT showed consistent performance regardless of types of QUDs, while GPT-2 encountered processing difficulties since a certain type of QUD required pragmatic inference for implicature. The findings revealed that, in terms of theoretical approaches, BERT inherently incorporates pragmatic implicature *not all* within the term *some*, adhering to Default model (Levinson, 2000). In contrast, GPT-2 seems to encounter processing difficulties in inferring pragmatic implicature within context, consistent with Context-driven model (Sperber and Wilson, 2002).

1 Introduction

In recent years, there has been remarkable progress in Natural Language Processing (NLP) thanks to the advent of Transformers (Vaswani et al., 2017), from which numerous Large Language Models (LLMs) have been developed. The effectiveness of these models relies on their ability to comprehend user input, which demands a focus on both semantics and pragmatics. Semantics involves the literal meanings of words or sentences, while pragmatics focuses on context-dependent intended meanings. Although advances in language

modeling, particularly in neural vector representations like Word2Vec (Mikolov et al., 2013) and GloVe (Pennington et al., 2014), have shown significant progress in semantics, pragmatic inference has not received as much attention in NLP research, despite its importance for achieving increasingly natural conversations with users.

Pragmatic inference refers to the process of making inference by considering the contexts, intentions, and situations of language use. As a type of pragmatic inference, **implicature** is regarded as a linguistic phenomenon where the speaker conveys additional meaning or information that is not explicitly stated. One of the most commonly studied implicatures is **scalar implicature**, which indicates the quantity or range of a particular attribute, such as *some*. Logically and semantically, the term *some* means *at least one and possibly all*. But, in actual language use, *some* is not always interpreted in this manner. Pragmatically, *some* would lead the hearer to infer the meaning *not all*.

- (1) Some students passed the exam.

For example, the sentence in (1) might be recognized as not all students passed the exam rather than at least one (or two in this case) and possibly all of them did.

However, Roberts (2012) suggested that, in pragmatic discourse, whether *some* is interpreted semantically or pragmatically depends on the surrounding context, such as **Question Under Discussion (QUD)**. QUD refers to topics in a conversation that should be addressed with relevant responses at a later stage in communicative interaction (Roberts, 2004; 2012; Beaver and Clark, 2008).

- (2) A: Did all students pass the exam?
B: Some students passed the exam.

Considering a conversational exchange in the form of QUD as in (2), *some* is more clearly interpreted as *not all* due to A’s question. This illustrates that *some* and *all* are positioned together or mutually related on an informational scale as $\langle \text{some}, \text{all} \rangle$, on which the less informative or weaker term *some* implies the negation of the more informative or stronger term *all* (Horn, 1972).

Several studies have attempted to explore whether LLMs can learn scalar implicature through Natural Language Inference (NLI) tasks (Jeretic et al., 2020; Schuster et al., 2020; Li et al., 2021). However, to our knowledge, the effects of manipulating context on scalar implicature have not been explored. Therefore, this study aims to investigate whether LLMs lean towards a semantic or pragmatic interpretation of scalar implicature and whether the interpretation can be influenced by context, drawing insights from experiments conducted in human language processing.

2 Background

2.1 Interpretations of scalar implicature

The study of deriving scalar implicature for the quantifier *some* has been widely conducted to investigate how pragmatically enriched meanings are computed. For example, the utterance in (3) semantically entails that *at least two and possibly all students passed the exam*, while pragmatically it is interpreted as *not all students passed the exam*, in which the meaning is enriched by the implicature (Geurts and Nouwen, 2007; Cummins and Katsos, 2010; Geurts et al., 2010).

- (3) a. *Utterance:*
Some students passed the exam.
b. *Semantic entailment:*
At least two and possibly all students passed the exam.
c. *Pragmatic implicature:*
Not all students passed the exam.

These two interpretations differ in whether *all* is negated or not, allowing for the possibility that *all* may still be valid in semantic interpretation. Furthermore, as shown in (4), the semantic entailment *at least one and possibly all* is not cancellable, while the interpretation of pragmatic implicature *not all* is cancellable (Grice, 1989; Geurts, 2010).

- (4) a. *Non-cancellable semantic entailment:*
Some students passed the exam. #In fact, none of them did.
b. *Cancellable pragmatic implicature:*
Some students passed the exam. In fact, all of them did.

This leads to the argument that *some* is positioned on a quantifier scale with varying levels of informativeness, ranging from the least to the most informative, representing the continuum $\langle \text{some}, \text{all} \rangle$ (Horn, 1972). The informativeness on the quantifier scale corresponds to the scale strength, where the less informative items are relatively weaker while the more informative ones are relatively stronger on the scale.

It is also argued that the hearer generally infers the speaker’s intention not to use the strong item (i.e., *all*) when trying to convey the meaning of the weak item (i.e., *some*). This is because interlocutors in conversation often expect that the speaker’s utterance would be optimally informative, as generalized by Gricean maxims (Grice, 1975). Therefore, scalar implicature leads to the general perception that the weak term implies the negation of the strong term on the scale.

However, *some* is not always interpreted with *not all* implicature. Roberts (2004) and Chierchia et al. (2012) have shown that the interpretation of *some* is heavily dependent on the broader context. Specifically, Roberts (2012) argued that whether *some* is interpreted with the pragmatic implicature is determined by the QUD, which refers to the topics in conversation that are expected to be addressed by appropriate answers (Roberts, 2004; 2012; Beaver and Clark, 2008). Examples can be found in (5) and (6), where the utterances containing *some* occur in response to different questions. The QUD that contains the term *all* is regarded as upper-bound as in (5), while the QUD that contain *any* is regarded as lower-bound as in (6).

- (5) *Upper-bound QUD:*
A: Did all students pass the exam?
B: Some students passed the exam.
(6) *Lower-bound QUD:*
A: Did any students pass the exam?
B: Some students passed the exam.

In the upper-bound QUD, the utterance of B is clearly interpreted as *not all students passed the exam*, suggesting *not all* implicature. On the other hand, the utterance of B in the lower-bound QUD can be felicitously interpreted, without *not all* implicature, as *at least two and possibly all students passed the exam*. The distinct interpretations of the same utterance in (5) and (6) arise due to the different questions asked by the speaker A. This illustrates that the utterance containing *some* may be ambiguous without any context, whereas a contextual cue, such as the QUD, can disambiguate the optimal interpretation of *some* in the discourse.

2.2 The processing of scalar implicature

Many studies have experimentally investigated whether scalar implicature is interpreted in semantic or pragmatic manner. For example, Bott and Noveck (2004) asked participants to judge the sentence in (7) is true or false.

(7) Some elephants are mammals.

Based on world knowledge, if *some* was interpreted semantically as *at least one and possibly all*, this sentence would be true; however, if *some* was interpreted pragmatically as *not all*, this sentence would be false. As a result, more participants judged these kinds of sentences as false, indicating a preference for pragmatic interpretation rather than semantic interpretation when scalar implicature was presented without context. These results have consistently appeared in other studies (Noveck and Posada, 2003; De Neys and Schaeken, 2007; Huang and Snedeker, 2009; Hunt et al., 2013; Tomlinson et al., 2013).

There have been two approaches to explain the processing of scalar implicature: Default model (Levinson, 2000) and Context-driven model (Wilson and Sperber, 1995; Sperber and Wilson, 2002). Levinson (2000) suggested, from the perspective of the Default model, that the hearer generally has an expectation of how language is typically used. This leads to *not all* implicature by default when encountering the term *some*. That is, implicature is generated as a default and can be negated or canceled when it becomes irrelevant in the given context. In contrast, Sperber and Wilson (2002) argued that scalar implicature is processed based on Relevance Theory. According to Relevance Theory, human cognition is generally

inclined to maximize relevance (Wilson and Sperber, 1995). This inclination allows a given utterance to be integrated with context, resulting in more positive cognitive effects for a more relevant utterance, while requiring greater processing effort for a less relevant utterance. In this view, the context plays a crucial role in determining whether the implicature is generated in the first place.

To examine the impact of context in the processing of scalar implicature, several studies have incorporated QUD in their experiments (Breheny et al., 2006; Zondervan et al., 2008; Politzer-Ahles and Fiorentino, 2013; Degen and Goodman, 2014; Dupuy et al., 2016; Politzer-Ahles and Husband, 2018; Yang et al., 2018; Ronai and Xiang, 2020).

(8) A: Did you fold all/any sweaters?
B: I folded some sweaters.

For example, Yang et al. (2018) presented participants with a situation where sentences containing *some* were followed by questions, as in (8). The QUD including *all* in the question is relevant to pragmatic implicature (i.e., *not all*), whereas the QUD including *any* in the question does not require implicature to interpret the conversation. In terms of pragmatic implicature, weak item *some* carries the meaning of negating the strong item *all*. Thus, if pragmatic implicature is appropriately established, the ratings for the sentences containing *some* should be lower, and the cognitive efforts required to infer the implicature should be greater in the *all*-condition than those in the *any*-condition. The experimental results exhibited that *all*-condition was rated lower than *any*-condition, suggesting that the interpretations of scalar implicature are sensitive to the given context. In addition, cognitive efforts measured in this study were greater when interpreting *some* in the upper-bound QUD (i.e., *all*-condition). This finding supports Context-driven model (Wilson and Sperber, 1995; Sperber and Wilson, 2002), indicating that more cognitive effort is required to derive scalar implicature.

Drawing from studies of human language processing related to scalar implicature, the current study poses the following questions regarding the language processing abilities of LLMs: